

Investment Decision Theory and Application- ECO4102Z/ECO4122Z

APT and Factor models

What is asset pricing?



Asset pricing theory provides a framework for **valuing assets** which generate an **uncertain stream of cash flows** in an **efficient market**. The key model in asset pricing theory has, for many years, been the CAPM.

However, the CAPM model has performed poorly in empirical tests.

Anomalies found

One major challenge to the CAPM was the set of results suggesting that it is possible to use knowledge of certain firm or security characteristics to develop profitable trading strategies, even after adjusting for investment risk as measured by beta.

Banz (1981) showed that portfolios of stocks with low market capitalizations (i.e., "small" stocks) outperformed "large" stock portfolios on a risk-adjusted basis,

Basu (1977) documented that stocks with low price earnings P/E ratios similarly outperformed high P/E stocks.

Fama and French (1992) also demonstrated that "value" stocks (i.e., those with high book value to- market price ratios) tend to produce larger risk-adjusted returns than "growth" stocks (i.e., those with low book-to-market ratios).

In an efficient market, these return differentials should not occur, meaning that either:

- (1) markets are not particularly efficient for extended periods of time (i.e., investors have been ignoring profitable investment opportunities for decades), or
- (2) market prices are efficient, but there is something wrong with the way the single-factor models such as the CAPM measure risk.

Factor Models

Factor model

A **factor model** is a financial model that explains the returns of assets (such as stocks, bonds, or portfolios) through exposure to various underlying risk factors.

The basic idea is that an asset's return is not only determined by idiosyncratic or asset-specific events but also by common factors that affect a broad range of assets in the market.

Factor models are used extensively in finance to understand, analyze, and predict asset returns and to construct diversified portfolios with specific risk and return characteristics.

Building a single factor model

The rate of return on any security can be decomposed into the sum of its expected + unanticipated components:

surprise/unexpected

$$r_i = E(r_i) + e_i$$

Suppose there is a common factor, m , which refers to a macroeconomic variable that affects all firms. Therefore, uncertainty can be decomposed into: uncertainty about the economy as a whole (m), and uncertainty about the firm in particular (e_i). Therefore:

$$r_i = E(r_i) + m + e_i$$

Building a single factor model

Since m and e_i are uncorrelated, the variance of share i will arise from two uncorrelated sources: systematic and firm specific. Therefore:

$$\sigma_i^2 = \overset{\text{mkt}}{\sigma_m^2} + \underset{\text{firm}}{\sigma^2(e_i)}$$

And the covariance between any two assets e_i and e_j will be:

$$\text{Cov}(r_i, r_j) = \text{Cov}(m + \overset{\text{firm}}{e_i}, m + e_j) = \sigma_m^2$$

*e cancel each other
because of diversifiable.*

Single Index Model (SIM)

However, some securities will be more sensitive than others to macroeconomic shocks (m). This can be captured by assigning a sensitivity coefficient to the macroeconomic variable. Therefore, the return on an asset is:

$$r_i = E(r_i) + \beta_i m + e_i$$

If we say 'm' can be proxied by a broad-based market index, we get the SIM:

$$R_i = \alpha_i + \beta_i R_M + e_i$$

Variance: $\sigma_i^2 = \overset{mkt}{\beta_i^2 \sigma_m^2} + \sigma^2(e_i)$

Unsystematic risk

Covariance: $\sigma_i \sigma_j \beta_i \beta_j \sigma_m^2$

Systematic risk

$$Cov(r_i, r_j) = Cov(\beta_i m + e_i, \beta_j m + e_j) = \beta_i \beta_j \sigma_m^2$$

Firm-specific risks are uncorrelated

Single-index model

If we replace 'm' with a broad-based market index, we get the SIM:

$$R_i = \alpha_i + \beta_i R_M + e_i$$

The SIM is used to reduce the complexity of large portfolio management by focusing on a single factor to explain returns (therefore we do not rely on the covariance estimates that Markowitz Portfolio Theory relies on).

The single index model is a crucial mechanism to other portfolio optimisation techniques (such as the Treynor Black model). We will cover this later in the semester.

Multi-factor models

- Multi-factor models describe returns as a function of a number of factors
- They are categorized into three
 - ① Macroeconomic-factor models
 - ② Fundamental-factor (or style-based) models
 - ③ Statistical-factor models

Output from factor models =

Factor loadings (sensitivities):

These are coefficients that measure how sensitive an asset is to each factor. For example, a stock with a high sensitivity to the market factor would tend to rise and fall with market movements.

Idiosyncratic (Specific) Risk: This is the component of an asset's return that cannot be explained by the factors. It's unique to the asset and unrelated to systematic factors. Idiosyncratic risk is assumed to be diversifiable in a well-constructed portfolio.

Macroeconomic Factor Models

Use observable economic time series as measures of the pervasive factors in stock returns.

The random return of a stock is therefore assumed to be linearly related to the macroeconomic shocks.



Economic growth

Exposure to the business cycle



Real rates

The risk of interest-rate movements



Inflation

Exposure to changes in prices



Credit

Default risk from lending to companies



Emerging markets

Political and sovereign risks



Liquidity

Holding illiquid assets

Fundamental (Style-based) factors

Rely on empirical findings that company attributes like size, dividend yield, book to market ratio, and industry classification explain a substantial portion of returns.



Value

Stocks discounted relative to their fundamentals



Minimum volatility

Stable, lower-risk stocks



Momentum

Stocks with upward price trends



Quality

Financially healthy companies
e.g. earnings yield



Size

Smaller, high-growth companies
mkt cap



Carry

Income incentive to hold riskier securities

Statistical factor models

A **statistical factor model** is a type of factor model in which the factors are not chosen based on economic theory or predefined characteristics (like size or value) but are instead derived from statistical methods applied to historical return data.

Statistical factor models use techniques like **Principal Component Analysis (PCA)** or **Factor Analysis** to identify the main sources of return variation within a dataset. These factors often lack a clear economic interpretation, as they are purely data-driven and reveal latent patterns or structures in the returns.

Macroeconomic Factor Models: Arbitrage Pricing Theory

Arbitrage Pricing Theory (APT) *Theoretical model*

*use arbitrage to eliminate mispricings
↳ net of 2 positions = 0*

Pricing approach which determines asset value based on Law of One price and no arbitrage

Assumptions of APT

- Asset returns are driven by multiple risk factors
- Linear relationship between returns and factors
- Large number of securities- all securities have finite expected values and variances, thus possible to form portfolios that diversify the firm specific risk of individual stocks
- Well-functioning markets do not allow for the persistence of arbitrage opportunities.

*• doesn't specify which factors to include
• doesn't specify how many factors to use*

CAPM equilibrium vs APT equilibrium

The CAPM is a risk-return dominance argument, implying that all investors hold mean-variance efficient portfolios. If a security is mispriced, then investors will tilt their portfolios toward the underpriced and away from the overpriced securities. Pressure on equilibrium prices results from many investors shifting their portfolios, each by a relatively small dollar amount.

With APT, the implication of a no-arbitrage condition is that a few investors who identify an arbitrage opportunity will mobilize large dollar amounts and quickly restore equilibrium.

Arbitrage Pricing Theory

APT can be characterised by two equations

$$R_{it} = a_i + \sum_{j=1}^k b_{ij} F_{jt} + \varepsilon_{it} \quad (1)$$

where the b_{ij} are known as *factor weights*.

the factors F are common across *all* securities and measure systematic risk.

$$ER_{it} = \lambda_0 + \sum_{j=1}^k b_{ij} \lambda_j \quad (2)$$

λ_0 = the expected return on an asset with zero systematic risk

λ_j = the risk premium related to the j th common risk factor

b_{ij} = the pricing relationship between the risk premium and the asset; that is, how responsive asset i is to the j th common factor. (These are called factor betas or factor loadings.)

Arbitrage Pricing Theory

APT can be characterised by two equations

$$R_{it} = a_i + \sum_{j=1}^k b_{ij} F_{jt} + \varepsilon_{it} \quad (1)$$

$$ER_{it} = \lambda_0 + \sum_{j=1}^k b_{ij} \lambda_j \quad (2)$$

Equation 1 shows that although each security is affected by all the factors

Equation 2 shows that the impact of any particular F_j depends on the value of b_{ij} and this is different for each security. This is the source of the covariance between the returns R_{it} on different securities

The APT predicts that 'general news' will affect the rate of return on *all* stocks but by different amounts. For example, a 1% unexpected rise in interest rates might affect the return on stocks of a company that was highly geared more than that for a company that was less geared.

Comparing CAPM vs APT

APT does not identify the risk factors and:

- (i) The relevant factors for a subset of securities may not be appropriate across different types of securities, and
- (ii) The relevant factors may change over time as market conditions evolve

Similarities to the CAPM: risk is only measured as systematic risk as unsystematic risk is assumed to be diversified away. There is no compensation for firm-specific risk.

	CAPM	APT
Form of Equation	Linear	Linear
Number of Risk Factors	1	$K (\geq 1)$
Factor Risk Premium	$[E(R_M) - RFR]$	$\{\lambda_j\}$
Factor Risk Sensitivity	β_i	$\{b_{ij}\}$
"Zero-Beta" Return	RFR	λ_0

APT example 1:

Consider the following example of two stocks and a two-factor model, with these risk factor definitions and sensitivities: *only include unanticipated Δs*

- anticipated are already priced*
- $\delta_1 =$ ~~unanticipated~~ *factor 1* changes in the rate of inflation. The risk premium related to this factor is 2 percent for every 1 percent change in the rate ($\lambda_1 = 0.02$). *λ_1*
- $\delta_2 =$ unexpected changes in the growth rate of real GDP. The average risk premium related to this factor is 3 percent for every 1 percent change in the rate of growth ($\lambda_2 = 0.03$). *factor 2*
- $\lambda_0 =$ the rate of return on a zero-systematic risk asset (i.e., zero-beta) is 4 percent ($\lambda_0 = 0.04$). *- risk free rate*

Assume also that there are two assets (x and y) that have the following sensitivities to these common risk factors:

- $\rightarrow$ sensitivity of x to factor 1*
- $b_{x1} =$ the response of asset x to changes in the inflation factor is 0.50
- $b_{x2} =$ the response of asset x to changes in the GDP factor is 1.50
- $b_{y1} =$ the response of asset y to changes in the inflation factor is 2.00
- $b_{y2} =$ the response of asset y to changes in the GDP factor is 1.75

APT example 1: Solution

δ_1 = unanticipated changes in the rate of inflation. The risk premium related to this factor is 2 percent for every 1 percent change in the rate ($\lambda_1 = 0.02$).

δ_2 = unexpected changes in the growth rate of real GDP. The average risk premium related to this factor is 3 percent for every 1 percent change in the rate of growth ($\lambda_2 = 0.03$).

λ_0 = the rate of return on a zero-systematic risk asset (i.e., zero-beta) is 4 percent ($\lambda_0 = 0.04$).

The overall expected return equation will be:

$$\begin{aligned} E(R_i) &= \lambda_0 + \lambda_1 b_{i1} + \lambda_2 b_{i2} \\ &= 0.04 + (0.02)b_{i1} + (0.03)b_{i2} \end{aligned}$$

APT example 1: Solution

b_{x1} = the response of asset x to changes in the inflation factor is 0.50

b_{x2} = the response of asset x to changes in the GDP factor is 1.50

b_{y1} = the response of asset y to changes in the inflation factor is 2.00

b_{y2} = the response of asset y to changes in the GDP factor is 1.75

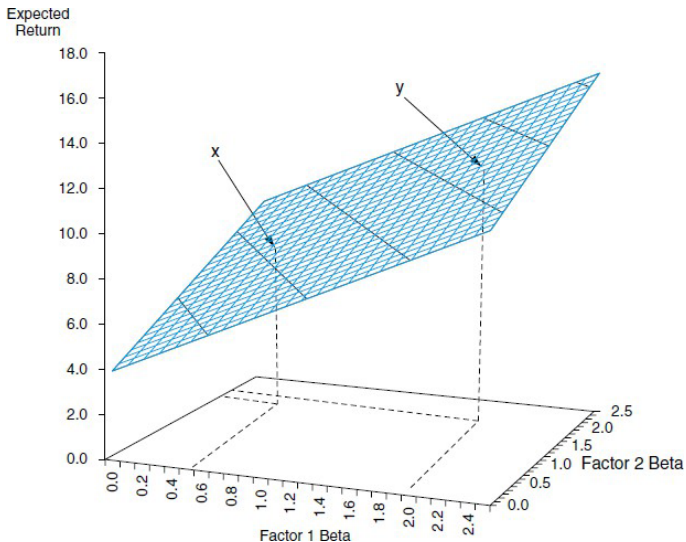
The returns of each of the two assets will be:

$$\begin{aligned} E(R_x) &= 0.04 + (0.02)(0.50) + (0.03)(1.50) \\ &= 0.0950 = 9.50\% \end{aligned}$$

$$\begin{aligned} E(R_y) &= 0.04 + (0.02)(2.00) + (0.03)(1.75) \\ &= 0.1325 = 13.25\% \end{aligned}$$

If the prices of the two assets do not reflect these expected returns, we would expect investors to enter into arbitrage arrangements selling overpriced assets short and using the proceeds to purchase the underpriced assets until the relevant prices are corrected.

SML under APT- Example two-factor SML



APT and Valuation- Example

- Suppose you have identified two common risk factors that affect asset returns in the South African financial market landscape, i.e. changes in GDP and changes in inflation. You have been evaluating the stocks of three firms (X, Y, Z) with the following relationship;

$$E(R_X) = 1.1\lambda_1 + 0.8\lambda_2$$

$$E(R_Y) = 0.7\lambda_1 + 0.6\lambda_2$$

$$E(R_Z) = 0.3\lambda_1 + 0.4\lambda_2$$

- If $\lambda_1 = 4\%$ and $\lambda_2 = 2\%$. Find the expected price of the three stocks, assuming none of them will pay a dividend and they have the same current price of $R30$.
- Suppose you know that the price of each stock next year would be $R31.50$, $R35.00$ and $R30.50$; respectively. Show how you can create a risk-less arbitrage investment by taking advantage of this mispricing. Assume short sales are allowed.

Suggested Solution

- Because the stock pays no dividend, all return is due to price appreciation.

$$\begin{aligned} E(R_x) &= 1.1 \times 0.04 + 0.8 \times 0.02 \\ &= 0.044 + 0.016 = 0.06 \text{ or } 6\% \end{aligned}$$

$$E(\text{Price } X) = 30(1.06) = 31.80$$

← *∴ short*

Price (1) =
31.50

$$\begin{aligned} E(R_y) &= 0.7 \times 0.04 + 0.6 \times 0.02 \\ &= 0.28 + .012 = 0.04 \text{ or } 4\% \end{aligned}$$

$$E(\text{Price } Y) = 30(1.04) = 31.20$$

Price (1) =
35.00

∴ long

$$\begin{aligned} E(R_z) &= 0.3 \times 0.04 + 0.4 \times 0.02 \\ &= 0.12 + 0.008 = 0.02 \text{ or } 2\% \end{aligned}$$

$$E(\text{Price } Z) = 30(1.02) = 30.60$$

Price (1) =
30.50

∴ short

Riskless arbitrage

The idea of riskless arbitrage is to assemble a portfolio that:

- (1) requires no net wealth invested initially and
- (2) will bear no systematic or unsystematic risk,
- (3) but (3) still earns a profit.
- (4) Letting w_i represent the percentage investment in Security i , the conditions that must be satisfied can be written formally as follows:

1. $\sum_i w_i = 0$ [i.e., no net wealth invested]
2. $\sum_i w_i b_{ij} = 0$ for all K factors [i.e., no systematic risk] and w_i is “small” for all i [i.e., unsystematic risk is fully diversified]
3. $\sum_i w_i R_i > 0$ [i.e., the actual portfolio return is positive]

Suggested solution continued:

- To create a riskless arbitrage investment, an investor would short 1 share of X and 1 of Z, and buy 2 of Y; with portfolio weights:
 $W_x = -0.5$, $W_y = +1.0$, and $W_z = -0.5$. The net investment is:

Short 1 share A = +30

Buy 2 shares B = -60

Short 1 share C = +30

Net investment = 0

- The risk exposure is:

Risk Exposure	Factor 1	Factor 2
X	$(-0.5) \times 1.1$	$(-0.5) \times 0.8$
Y	$(+1.0) \times 0.7$	$(+1.0) \times 0.6$
Z	$(-0.5) \times 0.3$	$(-0.5) \times 0.4$
Net Risk Exposure	0	0

or -1 and +2

- At the end of the period the profit is given by:

$$\begin{aligned} \text{Profit} &= (30 - 31.50) + 2(35 - 30) + (30 - 30.50) \\ &= -1.50 + 10 - 0.50 = 8 \end{aligned}$$

incurred loss of X + Y to get capital to buy Y

How can we test the APT?

Pre-specify factors: drawn from theory *but not clear how many factors are sufficient.*

- International studies suggest that **unexpected changes in industrial production, the term structure and yield spread are important.**
- South African studies suggest unanticipated changes in the DJIA, term structure of interest rates, short-term interest rates and a residual market factor are important.

Factor analysis: describes the variability among observed, correlated variables (i.e. the returns on a set of portfolios), in terms of a potentially lower number of unobserved variables called **factors**.

- E.g. It is possible that variations in 20 stock portfolios mainly reflect the variation in 2 unobserved variables.
- Limitations: subjectivity in determining the optimal no. of factors and no meaning can be assigned to the risk factors. However, it is possible to overcome this second limitation by comparing the risk factors to macroeconomic variables.

Two factor APT for SA

Van Rensburg and Slaney (1997) proposed the use of a **two-factor APT** model for asset pricing on the JSE with the **gold and industrial indices** representing the two factors.

This reflects **the dichotomy** on the JSE – i.e. that stock returns are driven by two distinct sources (gold and industrials). The **ASLI** appears “...to represent an aggregation of two disparate sets of forces rather than being a homogenous single factor underlying security returns”.

Two-factor model was able to explain 12.4% more of the variation in individual share returns than the CAPM, as measured by adjusted R^2 .

With the reclassification of the JSE indices, van Rensburg (2002) repeated the same factor analysis and found that the **Resources index** and the **Financial and Industrial Index** best captured the two factors prevalent in South African share returns.

$$E(r_p) = r_F + \beta_{pRESI}E(R_{RESI}) + \beta_{pFINDI}E(R_{FINDI})$$

This was re-evaluated by Laird-Smith et al (2016) who suggest that we utilise the South African Financials Index (SAFI) and the South African Resources Index (SARI).

We must be able to interpret the output from these tests. Consider the following:

1. Results from a time-series test of the two-factor APT on 1 stock:

Results	
α_i	-0.06***
β_{iFINDI}	0.50**
β_{iRESI}	0.22
$\bar{R}_2$	0.15

*, ** and *** indicate significance at 10%, 5% and 1%, respectively.

What is your conclusion about this stock and the two-factor model?

B_{FINDI} : how the stock moves with the FINDI

B_{RESI} : how the stock moves with the RESI

- This stock moves with the FINDI but is not as risky as the FINDI. It does not move with the RESI. **What does this tell you about the likely identity of this stock?**

This is likely a financial or industrial stock given that it moves with the FINDI but not the RESI e.g. Standard Bank.

As the intercept (alpha) is significant in the time-series tests, it suggests that this model does not hold (i.e. the stock does not earn a return commensurate with its level of systematic risk). As soon as the intercept is significant (irrespective of the sign), this is the conclusion we can draw.

2. Results from a cross-sectional test of the two-factor APT based on 100 stocks:

Results	
γ_0	0.02**
γ_{FINDI}	0.04**
γ_{RESI}	0.02**
$\bar{R}_2$	0.06

Adjusted R-squared?

What is your conclusion about the two-factor APT?

The intercept is significant which suggests that the SML does not run through the origin which goes against the APT. However, the slope coefficients are **significant and positive** which is consistent with the view **that higher risk should be associated with a higher reward**. Therefore there is mixed evidence about the validity of the two-factor model from the cross-sectional test.

Fundamental factor models

Fama-French (1993) Three factor model

a theoretical model

FF devised a three-factor model with the return on the market portfolio (as per the CAPM) and two additional factors:

$$E(R_i) = \alpha_i + \beta_{iM}E(R_M) + \beta_{iSMB}E(R_{SMB}) + \beta_{iHML}E(R_{HML}) + e_{it}$$



Where:

size
effect

$E(R_{SMB})$ is the expected return on the **SMB (small minus big) portfolio**: the return on a portfolio of small stocks less the return on a portfolio of big stocks

value
effect

$E(R_{HML})$ is the expected return on the **HML (high minus low) portfolio**: the return on a portfolio of stocks with high B/M ratios (value stocks) less the return on a portfolio of stocks with low B/M ratios (growth stocks).

The beta coefficients show the sensitivity of a stock to the market risk premium, the size premium and the value premium (β_{iM} , β_{iSMB} and β_{iHML}).

Where did these two additional factors come from?

Anomalies

Anomalies are empirical patterns that defy explanation by standard asset pricing theories.

“An anomaly is a deviation from the presently accepted paradigms that is too widespread to be ignored, too systematic to be dismissed as random error and too fundamental to be accommodated by relaxing the normative system” (Tversky and Kahneman, 1986:252).

Most anomalies were initially identified in the U.S. markets but thereafter found in markets around the world, including South Africa (although some differences are observed). It is largely clear, however, that these anomalies are **pervasive**.

The presence of anomalies signals the rejection of the joint CAPM/EMH hypothesis. Thus markets are either inefficient and/or the CAPM does not adequately describe returns.

Anomalies can be classified into different categories.

Anomalies continued:

Fama and French observed the following two anomalies in stock prices:

Size anomaly: **small** firms, those with low mkt caps, outperform **large** firms, after controlling for beta (i.e. on a risk-adjusted basis). Recent evidence suggests that the size effect may be disappearing on the U.S. market.

Value anomaly: **value** firms, those with low ratios of price relative to their fundamentals (low P/E and high B/M ratios) outperform **growth** firms – firms which have high prices relative to their fundamentals, after adjusting for beta.

Therefore, the authors concluded that size and the B/M ratio can predict deviations of stock returns from levels consistent with the CAPM.

Expansion of the CAPM - market factor still included and expected to capture systematic risk arising from macroeconomic factors.

The model explains the size and value anomalies which the CAPM cannot (see next slide).

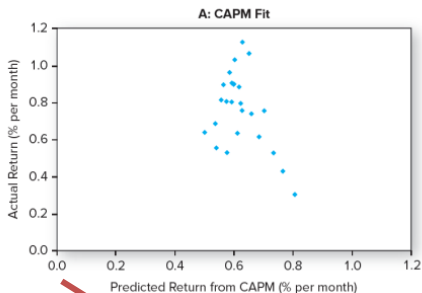
While SMB and HML are not themselves obvious candidates for relevant risk factors, they can be seen to **proxy for hard to measure fundamental variables**. E.g. small firms and those with high B/M ratios are more likely to be in financial distress; small stocks may be more sensitive to changes in business conditions; small stocks are more illiquid (size premium = illiquidity premium); size and B/M ratio seem to predict GDP growth and thus may capture some elements of business cycle risk.

The model has **no theoretical foundation**. Essentially, FF “backfit” their model to an APT framework.
And, it has some empirical shortcomings ...

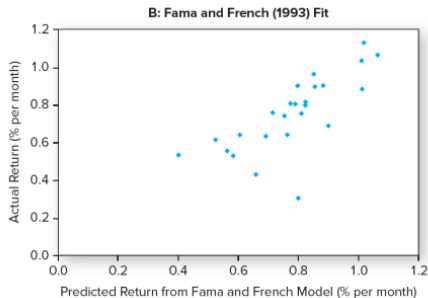
Does the model work?

Model works well internationally in explaining returns of portfolios sorted on size- and value *BUT cannot explain the momentum anomaly.*

JSE evidence: model performs well for South African stocks.



Predicted returns from the CAPM are far from actual returns



Predicted returns from the three-factor model are close to actual returns

We must be able to interpret the output from these tests. Consider the following:
** know what the factors are + where we get them e.g. SMB*

1. Results from a time-series test of the three-factor model on 1 stock:

Results	
α_i	-0.03
β_{iM}	0.43**
β_{iSMB}	-0.39*
β_{iHML}	0.19**
$\bar{R}_2$	0.23

*, ** and *** indicate significance at 10%, 5% and 1%, respectively.

What is your conclusion about this stock and the three-factor model?

B_M : interpretation identical to under the CAPM.

B_{SMB} : how the stock moves with the SMB portfolio returns

B_{HML} : how the stock moves with the HML portfolio returns

- Positive (negative) B_{SMB} = stock moves with returns of small (big) stocks
- Positive (negative) B_{HML} = stock moves with returns of value (growth) stocks
- What does this tell you about the likely identity of this stock?

As the intercept (alpha) is **insignificant** in the time-series tests, it suggests that this model does hold (i.e. the stock earns a return commensurate with its level of systematic risk).

This is a **defensive** (market beta is less than one), **large** (coefficient on SMB is negative) and **value** (coefficient on HML is positive) stock.

2. Results from a cross-sectional test of the three-factor model based on 100

Results	
γ_0	0.03
γ_M	0.06**
γ_{SMB}	0.01**
γ_{HML}	0.04**
$\bar{R}_2$	0.13

What is your conclusion about three-factor model?

The intercept is **insignificant** which suggests that the SML runs through the origin which is consistent with expectations. The **coefficients are all significant and positive** for the three factors which is consistent with the theory that **higher risk should be associated with a higher reward**. Therefore, there is evidence here that supports the three-factor model.

Carhart (1997) Four-Factor Model

As the three-factor model cannot explain the momentum anomaly, Carhart (1997) proposed a four-factor model:

$$E(R_i) = \alpha_i + \beta_{iM}E(R_M) + \beta_{iSMB}E(R_{SMB}) + \beta_{iHML}E(R_{HMLt}) + \beta_{iWML}E(R_{WMLt}) + e_{it}$$

where: $E(R_{WML})$ is the expected return on the **WML (winners minus losers) portfolio**: the return on a portfolio of past winners less the return on a portfolio of past losers.

Momentum anomaly: refers to short-term persistence in returns where shares which have performed well in the past one to 12 months continue to perform well up to a year thereafter, while those that have performed poorly continue to perform poorly.

Carhart (1997) Four-Factor Model

The model has become popular to evaluate the performance of mutual funds ^{unit trusts} i.e. use the alpha instead of the alpha from CAPM. ***

What does this factor capture as a source of risk?

Somewhat controversial ... some evidence that it may be related to industrial production or business cycles.

Momentum is seen to be explained more by behavioural biases than poor risk measurement.

Behavioural explanations:

The slow reaction to news by investors (*conservatism bias*) means that past performance is extrapolated giving rise to momentum in share returns.

Herding behavior can drive momentum, pushing prices higher in the short term.

Investors seek out information that confirms their existing beliefs (*confirmation bias*). After seeing a stock rise, they may become more confident in its continued success and ignore contrary signals, leading to momentum.

Fama & French (2015) Five-Factor Model *built from theory*

Investment anomaly: firms with **conservative** investment (low ratios of investment to total assets) outperform those with **aggressive** investment (investment relative to total assets), after adjusting for beta.

Profitability anomaly: firms with **robust** profitability (high ratios of earnings to total assets) outperform those with **weak** profitability (low ratios of earnings to total assets) after adjusting for beta.

Fama and French (2015) introduced a five-factor model which includes factors relating to the **profitability** and **level of investment** of a firm:

$$E(R_i) = \alpha_i + \beta_{iM}E(R_M) + \beta_{iSMB}E(R_{SMB}) + \beta_{iHML}E(R_{HMLt}) + \beta_{iCMA}E(R_{CMAt}) + \beta_{iRMW}E(R_{RMWt}) + e_{it}$$

Fama & French (2015) Five-Factor Model

Fama and French (2015) introduced a five-factor model which includes factors relating to the **profitability** and **level of investment** of a firm:

$$E(R_i) = \alpha_i + \beta_{iM}E(R_M) + \beta_{iSMB}E(R_{SMB}) + \beta_{iHML}E(R_{HMLt}) + \beta_{iCMA}E(R_{CMAit}) + \beta_{iRMW}E(R_{RMWt}) + e_{it}$$

where: $E(R_{CMA})$ is the expected return on the **CMA (conservative minus aggressive) portfolio**: the return on a portfolio of firms with conservative investment (low I/A ratios) less the return on a portfolio of stocks with aggressive investment (high I/A ratios) and $E(R_{RMW})$ is the expected return on the **RMW (robust minus weak) portfolio**: the return on a portfolio of firms with robust profitability (high E/A ratios) less the return on a portfolio of stocks with weak profitability (low E/A ratios).

In contrast to the three-factor model, which is based on empirical research and backfitted to the APT, the justification for the five-factor model stems from the **bottom-up DDM** - the expected stock return modelled by the DDM depends on the B/M ratio, expected earnings and expected growth in book equity. The inclusion of these two factors was also influenced by the findings of the investment and profitability anomalies.

Does the model work?

Except for momentum, the model reduces the anomalies left unexplained by the three-factor model (Fama & French, 2016).

JSE evidence: Five-factor model best explains stock returns.

Factors can be found on Kenneth French's [website](#).
South African data available from Peresec [website](#).

Pros and Cons of different models:

Method	Pros	Cons
Macroeconomic Variables	provides intuition	measurement difficulties
Fundamental factors	more intuitive than statistical	characteristics based on past anomalies
Statistical factors	provides no economic intuition	factors not unique assumes constant covariances/variances